

Practice Exam 2

① $X \sim N(1200, 200^2)$

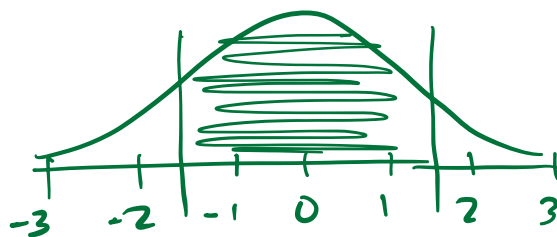
a) $P(900 < X < 1500)$

$$P\left(\frac{900-1200}{200} < \frac{X-\mu}{\sigma} < \frac{1500-1200}{200}\right) \quad \text{convert to z-scores}$$

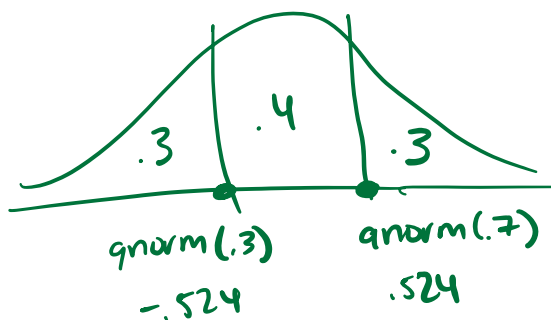
$$P\left(-\frac{3}{2} < Z < \frac{3}{2}\right)$$

$$\text{pnorm}(1.5) - \text{pnorm}(-1.5)$$

$$0.933 - 0.067 = \boxed{0.866}$$



b) middle 40%



$$Z = .524 = \frac{X - \mu}{\sigma} = \frac{X - 1200}{200}$$

solve for X

$$200(.524) + 1200 = X$$

$$1304.8$$

$$1200 - 200(.524) = 1095.2$$

$$[1095.2, 1304.8]$$

c) $X = \#$ of times sales are outside (\$900, \$1500) out of 5

$$X \sim \text{binomial}(n=5, p = 1 - \underset{\substack{\uparrow \\ \text{from part a}}}{.866})$$

$$P(X \geq 2) = 1 - P(X \leq 1) = 1 - P(X=0) - P(X=1)$$

$$= 1 - \binom{5}{0}(.134)^0(.866)^5 - \binom{5}{1}(.134)^1(.866)^4$$

$$= \boxed{0.1361}$$

2) a) Let $Y = \#$ defective

$Y \sim \text{binomial}(n=10, p=0.12)$

@ least 7 not defective means at most 3 defective

$$P(Y \leq 3) = P(Y=0) + P(Y=1) + P(Y=2) + P(Y=3)$$

$$= \binom{10}{0} \cdot .12^0 (.88)^{10} + \binom{10}{1} \cdot .12^1 (.88)^9 + \\ \binom{10}{2} \cdot .12^2 (.88)^8 + \binom{10}{3} \cdot .12^3 (.88)^7$$

$$= .2785 + .3798 + .2330 + .0847$$

$$= \boxed{.9761}$$

b) $Y = \#$ of trials until 1st defective

$Y \sim \text{geometric}(p=.12)$

$$P(Y=8) = .12 (.88)^7 = \boxed{.049}$$

c) still use geometric, but we want $P(Y=1 \text{ or } Y=3 \text{ or } Y=5 \dots)$

$$P(Y = \text{odd number}) = P(Y = 2k+1) \text{ where } k=0,1,\dots$$

$$= \sum_{k=0}^{\infty} .12 (.88)^{2k+1-1} = \sum_{k=0}^{\infty} .12 (.88^2)^k$$

geometric series $\rightarrow$ recall $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$

$$= .12 \left(\frac{1}{1-.88^2} \right)$$

$$= \boxed{0.5319}$$

$$③ \quad f(x) = \begin{cases} x^2 & -1 < x < 0 \\ cx & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$a) \quad 1 = \int_{-1}^0 x^2 dx + \int_0^2 cx dx$$

$$= \left. \frac{x^3}{3} \right|_{-1}^0 + \left. \frac{c}{2} x^2 \right|_0^2$$

$$1 = \frac{1}{3} + 2c$$

$$\frac{2}{3} = 2c \Rightarrow \boxed{c = \frac{1}{3}}$$

$$b) \quad F(x) = \begin{cases} \int_{-1}^x t^2 dt = \left. \frac{t^3}{3} \right|_{-1}^x = \frac{x^3}{3} + \frac{1}{3} & -1 < x < 0 \\ F(0) + \int_0^x \frac{1}{3} t dt = \frac{1}{6} t^2 \Big|_0^x = \frac{1}{6} x^2 & 0 < x < 2 \end{cases}$$

add this b/c its a cumulative probability

$$F(x) = \begin{cases} \frac{x^3+1}{3} & -1 < x < 0 \\ \frac{1}{3} + \frac{1}{6} x^2 & 0 < x < 2 \\ 1 & x > 2 \\ 0 & x < -1 \end{cases}$$

note these should return same values at $x=0$

④ $S \sim U[2, 4]$

a) $f(s) = \frac{1}{4-2} = \frac{1}{2} \quad 2 < s < 4$

b) $E(A) = E(S^2) \quad \text{use } V(s) = E(S^2) - (E(s))^2$

$$E(s) = \frac{2+4}{2} = 3$$

$$V(s) = \frac{(b-a)^2}{12} = \frac{(4-2)^2}{12} = \frac{4}{12} = \frac{1}{3}$$

properties of
uniform
dist.

$$E(S^2) = V(s) + (E(s))^2$$

$$= \frac{1}{3} + 3^2 = \boxed{\frac{28}{3}}$$

c) $V(S^2) = E(S^4) - (E(S^2))^2$

$$E(S^4) = \int_2^4 s^4 f(s) ds = \int_2^4 s^4 \frac{1}{2} ds = \frac{s^5}{5} \frac{1}{2} \Big|_2^4$$

$$= \frac{4^5 - 2^5}{10} = 99.2$$

$$V(S^2) = 99.2 - \left(\frac{28}{3}\right)^2 = \boxed{12.09}$$

5) $Y \sim \text{Poisson}(\lambda)$

a) $V(Y) = E(Y^2) - (E(Y))^2 \Rightarrow E(Y^2) = V(Y) + (E(Y))^2$
 we know $E(Y) = V(Y) = \lambda$, so $\boxed{E(Y^2) = \lambda + \lambda^2}$

b) $E[Y(Y-1)(Y-2)] = \sum_{y=0}^{\infty} \underbrace{y(y-1)(y-2)}_{\text{will be 0 for } y=0,1,2} \frac{\lambda^y e^{-\lambda}}{y!}$

$= \sum_{y=3}^{\infty} \frac{y(y-1)(y-2) \lambda^y e^{-\lambda}}{y!} = \sum_{y-3=0}^{\infty} \frac{\lambda^y e^{-\lambda}}{(y-3)!}$

$= e^{-\lambda} \sum_{y-3=0}^{\infty} \frac{\lambda^{y-3} \lambda^3}{(y-3)!} = e^{-\lambda} \lambda^3 e^{\lambda} = \boxed{\lambda^3}$
 $\parallel$
 e^{λ} by Maclaurin series

c) $\lambda^3 = E[Y(Y-1)(Y-2)] = E[Y(Y^2 - 3Y + 2)] = E[Y^3 - 3Y^2 + 2Y]$
 $= E(Y^3) - 3E(Y^2) + 2E(Y)$

$\Rightarrow E(Y^3) = \lambda^3 + 3E(Y^2) - 2E(Y)$
 $= \lambda^3 + 3(\lambda + \lambda^2) - 2\lambda$
 $= \boxed{\lambda^3 + 3\lambda^2 + \lambda}$

$$\textcircled{6} \quad m(t) = (0.5 + 0.5e^{2t})^2$$

$$m'(t) = 2(0.5 + 0.5e^{2t}) (e^{2t}) =$$

$$m'(0) = 2(0.5 + 0.5e^0) (e^0) \\ = 2(1)(1) = \boxed{2 = E(X)}$$

$$V(X) = m''(t=0) - (m'(t=0))^2$$

$$\text{Simplify } m'(t) = e^{2t} + e^{4t}$$

$$m''(t) = 2e^{2t} + 4e^{4t}$$

$$m''(0) = 2e^0 + 4e^0 = 6 = E(X^2)$$

$$V(X) = 6 - 2^2 = \boxed{2}$$

7 a) $f(y) = 2(1-y) \quad 0 \leq y \leq 1$

$$E(y) = \int y \cdot 2(1-y) dy = \int_0^1 2y - 2y^2 dy$$

$$= y^2 \Big|_0^1 - \frac{2}{3} y^3 \Big|_0^1 = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\frac{1}{3}(\$100,000) = \boxed{\$33,333.33} \text{ average claim}$$

b) $F(y) = 0.9$ solve for y

$$F(y) = \int_0^y 2(1-t) dt = 2 \left(t - \frac{t^2}{2} \right) \Big|_0^y$$

$$= 2 \left(y - \frac{y^2}{2} \right) = 2y - y^2$$

solve $0.9 = 2y - y^2$

$$y^2 - 2y + 0.9 = 0$$

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\frac{2 \pm \sqrt{4 - 4(1)(0.9)}}{2}$$

$$1 \pm \frac{\sqrt{.4}}{2}$$

$$1 \pm .316$$

$$\boxed{.684}$$

(the other solution 1.316 is not in support)

c) $P(Y < .75) = F(.75) = 2(.75) - .75^2$

↑
\$75,000

$$= \boxed{.9375}$$

⑧ can't be cdf b/c $h(150) \neq 1$ for $y > 150$
to check if pdf, see if it integrates to 1

$$\begin{aligned} 1 &\stackrel{?}{=} \int_{100}^{150} h(y) dy = \int_{100}^{150} (0.02y - 2) dy \\ &= \left. \frac{.02}{2} y^2 \right|_{100}^{150} - 2y \Big|_{100}^{150} \\ &= 125 - 2(150 - 100) \\ &= 100 \neq 1 \end{aligned}$$

⌈ ⑨ neither pdf or cdf ⌋